

Does spin-polarized-fuel neutron steering survive a real stellarator wall?

OpenMC transport of SPF DT neutrons through a conformal
breeding blanket on precise_QA ($nfp = 2$). Prepared for
discussion with E. Peterson (MIT PSFC).

T. Kiker, Columbia University

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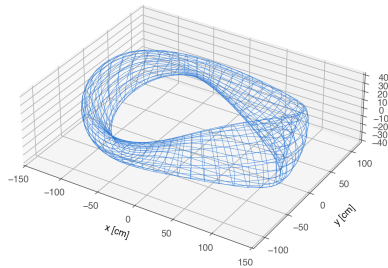
The question

- ▶ Spin-polarized fuel biases the DT neutron emission direction relative to the local field B .
- ▶ Idea: steer the fast-neutron load relative to the magnets.
- ▶ Two things could kill it — a strongly-shaped 3D wall, and scattering in a thick blanket.
- ▶ Machinery validation on a public equilibrium.

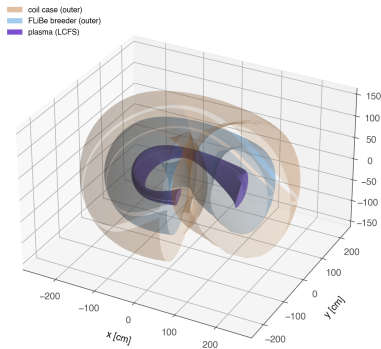
$$I(\theta) \propto \sin^2 \theta \quad (\perp) \quad \text{or} \quad 1 + 3 \cos^2 \theta \quad (\parallel), \quad \theta \text{ from } B$$

The device

Plasma boundary (LCFS), precise_QA, 2 field periods

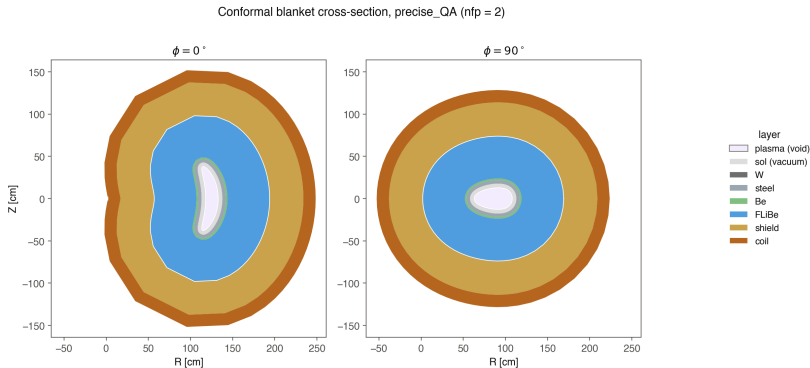


Conformal blanket wraps the plasma (300° cutaway)



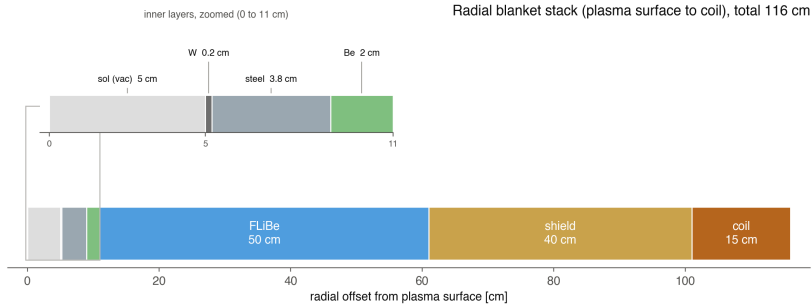
- precise_QA, $nfp = 2$; a conformal multi-layer blanket wraps the plasma.

Blanket cross-section



- ▶ Crescent plasma pushed outboard at $\varphi = 0$; rounder and centered at $\varphi = 90^\circ$.
- ▶ Offset shells become near-circular and de-center from the plasma.

Radial stack

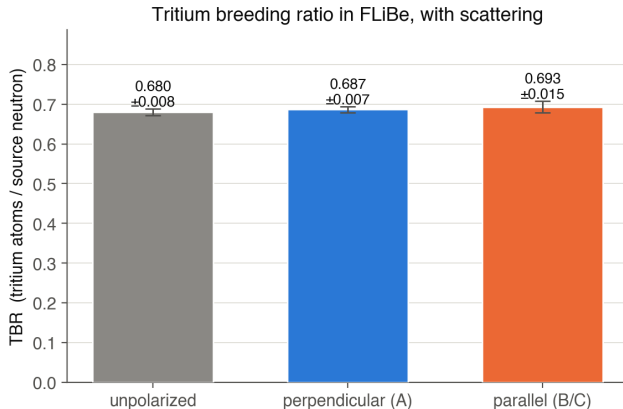


- ▶ The 116 cm layer stack, plasma surface outward.
- ▶ FLiBe is the 50 cm breeder and coolant.

The pipeline

- ▶ Polarized CompiledSource — sampler verified against the DT differential cross-section.
- ▶ Hand-built watertight DAGMC geometry to bypass a tooling blocker.
- ▶ OpenMC transport in two streams: free and scatter.
- ▶ Tallies: TBR, coil fast flux, per-layer heating, first-wall current mesh.

Tritium breeding ratio

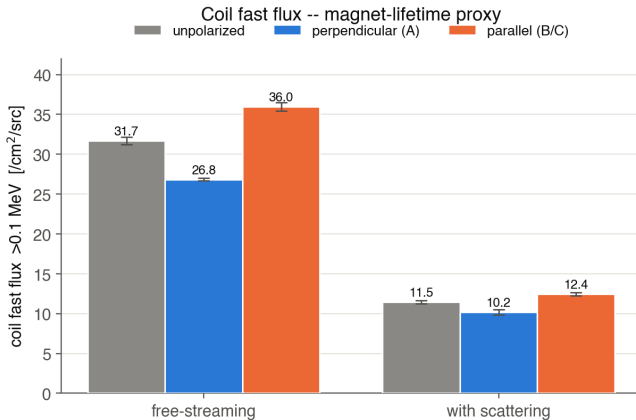


Free-streaming TBR ≈ 0 by construction (no scattering, so no moderation and no capture). 10k-history smoke run.

- ▶ 0.68–0.69 tritium per source neutron in the scatter stream.
- ▶ Polarization barely moves it, within about 1σ .
- ▶ Free-streaming TBR ≈ 0 by construction — no moderation.

10k-history smoke run — indicative.

Coil fast flux



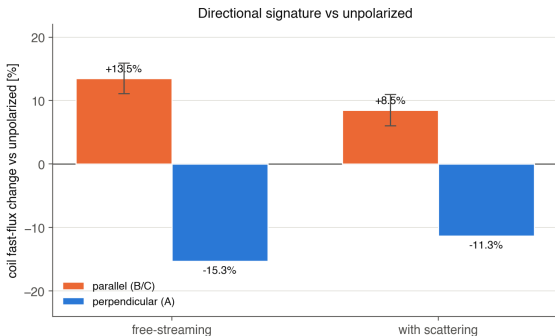
Parallel raises, perpendicular lowers the coil load; the ordering survives scattering. 10k-history smoke run ($\pm$ shown; large MC error at depth).

- ▶ Parallel raises the coil load; perpendicular lowers it.
- ▶ The ordering survives scattering.
- ▶ Swing about 34% free, about 22% with scattering.

10k-history smoke run — indicative.

Directional efficiency η

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Large Monte-Carlo error on a 10k-history smoke run ($\eta = 0.63 \pm 0.22$); indicative, needs variance reduction (weight windows / FW-CADIS) for a precise value.

$$\eta = \frac{+8.5\%}{+13.5\%}$$

η

0.63

± 0.22

fraction of the free-streaming steering signal that survives conformal walls + scattering

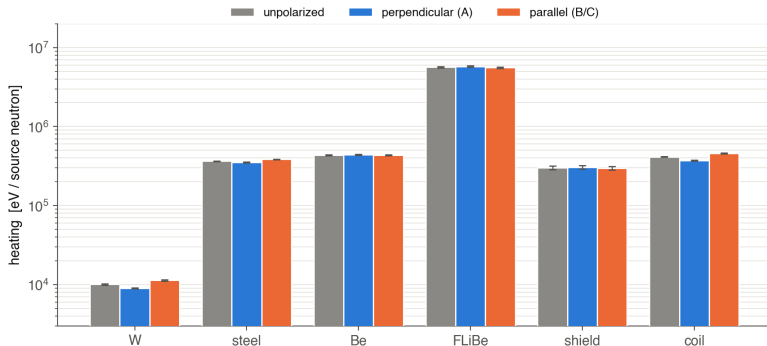
$$\eta \equiv \Delta_{\text{scatter}} / \Delta_{\text{free}} = 8.5\% / 13.5\% \approx 0.63 \pm 0.22$$

- ▶ Parallel coil flux: +13.5% free to +8.5% with scattering.
- ▶ η : the fraction of the free-streaming steering that survives.

10k-history smoke run — indicative

Per-layer heating

Per-layer nuclear heating, with scattering



FLiBe (50 cm breeder) absorbs most of the energy. At the coil, parallel > unpolarized > perpendicular -- same directional ordering as the coil flux. 10k-history smoke run.

- ▶ FLiBe absorbs most of the deposited energy.
- ▶ At the coil: parallel > unpolarized > perpendicular.
- ▶ Same sign as the coil flux — independent corroboration.

10k-history smoke run — indicative.

Caveats and what's next

- ▶ 10k-history smoke run; large MC error deep behind the blanket.
- ▶ Machinery validation, not a reactor claim.
- ▶ Naive constant-offset wall; small inboard self-fold at $\varphi = 0$ needs a proper offset.
- ▶ Next: variance reduction (weight windows / FW-CADIS) to converge η .
- ▶ Then an analytic wall-load cross-check, higher statistics, and physics equilibria.

Backup: raw numbers

mode	TBR (scatter)	flux (free) [cm^2/src]	flux (scatter) [cm^2/src]	coil heat [eV/src]
unpolarized	0.680 ± 0.008	31.68 ± 0.48	11.45 ± 0.19	$414,510 \pm 4,844$
perpendicular	0.687 ± 0.007	26.84 ± 0.19	10.16 ± 0.33	$371,994 \pm 4,856$
parallel	0.693 ± 0.015	35.96 ± 0.54	12.43 ± 0.20	$457,856 \pm 7,632$

Directional efficiency (parallel, scatter over free): $\eta = 0.63 \pm 0.22$.